



University of Central Punjab

Faculty of Information Technology
BS Software Engineering

Advanced Web Programming (Spring 2026)
Project Documentation

Net Zero — Expense Tracker App

Project Title	Net Zero — Expense Tracker App
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1. Project Overview

Net Zero is a mobile expense tracker application made using React Native and Expo. The main idea behind this app is to give users a simple way to keep track of their daily spending without any complicated setup or account creation. Users can just open the app and start adding their expenses right away.

The app is a mobile application that works on both Android and iOS. It stores all data locally on the device so there is no need for internet or any backend server. The target audience is mainly students and young professionals who want a quick and simple way to monitor their monthly spending.

2. Problem Statement

Most people do not track their expenses properly and end up spending more than they planned. The existing options are either too complex with too many features, or they require you to sign up and create an account just to use basic functionality.

Some common issues with existing apps are:

- They require internet connection and account registration even for basic use.
- Apps like bank clients have too many features that a regular user does not need.
- Manual tracking on paper or in a notes app is very inconvenient.
- Most apps do not show amounts in Pakistani Rupees (PKR) by default.

Net Zero solves this by being a simple, offline, no-login app where users can just open it and start logging their expenses instantly.

3. Project Objectives

- Allow users to add, edit, and delete expenses easily.
- Store all data locally on the phone using AsyncStorage.
- Let users filter expenses by month and by category.
- Show a spending summary with a pie chart so users can see where their money is going.
- Support 5 expense categories: Food, Transport, Shopping, Bills, and Other.
- Keep the UI clean and simple using a dark theme.
- Demonstrate React Native concepts like Context API, useReducer, FlatList, and React Navigation.

4. Scope of the Project

What is included in the project:

- Adding, editing, and deleting expense entries.
- Month selector to view expenses for any month.
- Category filter chips on the expenses screen (Food, Transport, Shopping, Bills, Other, All).
- Home dashboard showing total spent and recent transactions.
- Summary screen with a pie chart and category-wise breakdown.
- Data saved locally using AsyncStorage so it persists between app restarts.
- Dark themed UI with consistent colors throughout the app.

What is NOT included (may be added later):

- Cloud sync or multi-device support.
- User login or authentication.
- Budget limits and alerts.
- Exporting data as PDF or CSV.
- Income tracking.

5. Functional Requirements

These are the things the app will do:

Expense Management:

- User can tap a button to open an add expense form.
- The form has fields for title, amount, date, and category.
- User can tap any existing expense to edit it.
- User can delete an expense from the edit form or from the expenses list.

Month Filtering:

- User can navigate between months using arrow buttons.
- The selected month is saved so it stays the same when the app is reopened.
- All screens show data for the same selected month.

Summary and Dashboard:

- Home screen shows total PKR spent and number of transactions for the month.

- Summary screen shows a pie chart and a list with percentages per category.
- If there are no expenses for a month, an empty state message is shown.

6. Non-Functional Requirements

Usability:

- The app uses a consistent dark theme with a purple accent color throughout.
- All buttons and cards give visual feedback when tapped.
- Empty state messages guide the user when there is no data.

Performance:

- FlatList is used for expense lists so the app stays fast even with many entries.
- Data is loaded asynchronously at startup so the UI does not freeze.

Responsiveness:

- The layout works properly on different Android screen sizes.
- SafeAreaView is used to handle device notches properly.

Privacy and Security:

- All data stays on the device. Nothing is sent to any server.
- No permissions like camera, contacts, or location are requested.
- No login required so there is no risk of credential issues.

7. Technology Stack

Frontend Framework: React Native 0.81.5 with Expo ~54.0.34

Language: JavaScript (ES6+)

Navigation: @react-navigation/native and @react-navigation/bottom-tabs

Local Storage: @react-native-async-storage/async-storage

Charts: react-native-chart-kit with react-native-svg

Icons: @expo/vector-icons (Ionicons)

Date Utilities: date-fns

Other: react-native-safe-area-context, react-native-screens

8. Application Flow / Architecture

The app is organized into a src folder with subfolders for screens, components, context, utils, constants, and navigation.

When the app starts, the ExpensesProvider loads all saved expenses and the last selected month from AsyncStorage. This data is made available to all screens through React Context API so any screen can access or update the expenses without passing props manually.

The state is managed using useReducer inside the context. Actions like ADD, UPDATE, DELETE, and SET_MONTH are dispatched when the user does something. Whenever the expenses array changes, a useEffect automatically saves the updated data back to AsyncStorage.

Navigation is handled by a bottom tab navigator with three tabs: Home, Expenses, and Summary. All three screens share the same selected month through context, so switching tabs always shows the data for the same month.

The ExpenseModal component is a modal overlay that is used on both the Home screen and the Expenses screen. When opened for a new expense it shows an empty form. When opened from an existing expense card it pre-fills the form with that expense's data and shows a delete button.

9. UI Screens / Components Description

The app has 3 main screens and 1 modal. All screens use a dark background (#0D0D0D) with purple (#7C5AF6) as the primary accent color.

9.1 Home Screen

This is the first screen the user sees. Its purpose is to give a quick overview of monthly spending.

- App name "Net Zero" shown at the top.
- A hero card showing total PKR spent and number of transactions.
- Month Selector component with left/right arrows to change month.
- A "Recent Transactions" section showing the last 20 expenses.
- An "Add" button to open the expense form.
- Tapping any expense card opens it for editing.
- If no expenses exist, an empty state with icon and message is shown.

9.2 Expenses Screen

This screen shows the full list of expenses for the selected month with filtering options.

- Month Selector at the top (same as home screen).
- Horizontal scrollable filter chips: All, Food, Transport, Shopping, Bills, Other.
- Active filter chip is highlighted with the category color.
- Shows how many expenses are currently displayed.
- Full scrollable list of filtered expenses using FlatList.
- Each card has a delete button on the right side.
- Tapping a card opens the edit modal.

9.3 Summary Screen

This screen gives a visual breakdown of spending by category.

- Month Selector at the top.
- A card showing the total amount spent for the month.
- A pie chart showing spending split by category with color-coded slices.
- A list below showing each category with its name, percentage, a progress bar, and PKR amount.
- Empty state shown if no expenses for that month.

9.4 Expense Modal (Add / Edit)

This is a modal overlay used for adding new expenses or editing existing ones.

- Title text field.
- Amount field with numeric keyboard.
- Date picker field.
- Category selector with chips for Food, Transport, Shopping, Bills, Other.
- Save button to add or update the expense.
- Delete button (only shown in edit mode) to remove the expense.
- Close/cancel button to dismiss without saving.

10. API Usage

Net Zero does not use any external APIs or web services. The app is completely offline. Data is stored and retrieved using the AsyncStorage library from [@react-native-async-storage/async-storage](https://react-native-async-storage.github.io/async-storage/).

There are utility functions in `src/utils/storage.js` that handle all storage operations:

- `loadExpenses()` – reads the saved expenses from `AsyncStorage` and returns them as an array.
- `saveExpenses(expenses)` – saves the full expenses array to `AsyncStorage` as JSON.
- `generateId()` – creates a unique ID for each new expense using the current timestamp.
- `loadSelectedDate()` – loads the last selected month and year so the app can restore it.
- `saveSelectedDate(month, year)` – saves the selected month whenever the user changes it.

Each expense entry is stored in this format:

```
{ id, title, amount, category, date, createdAt }
```

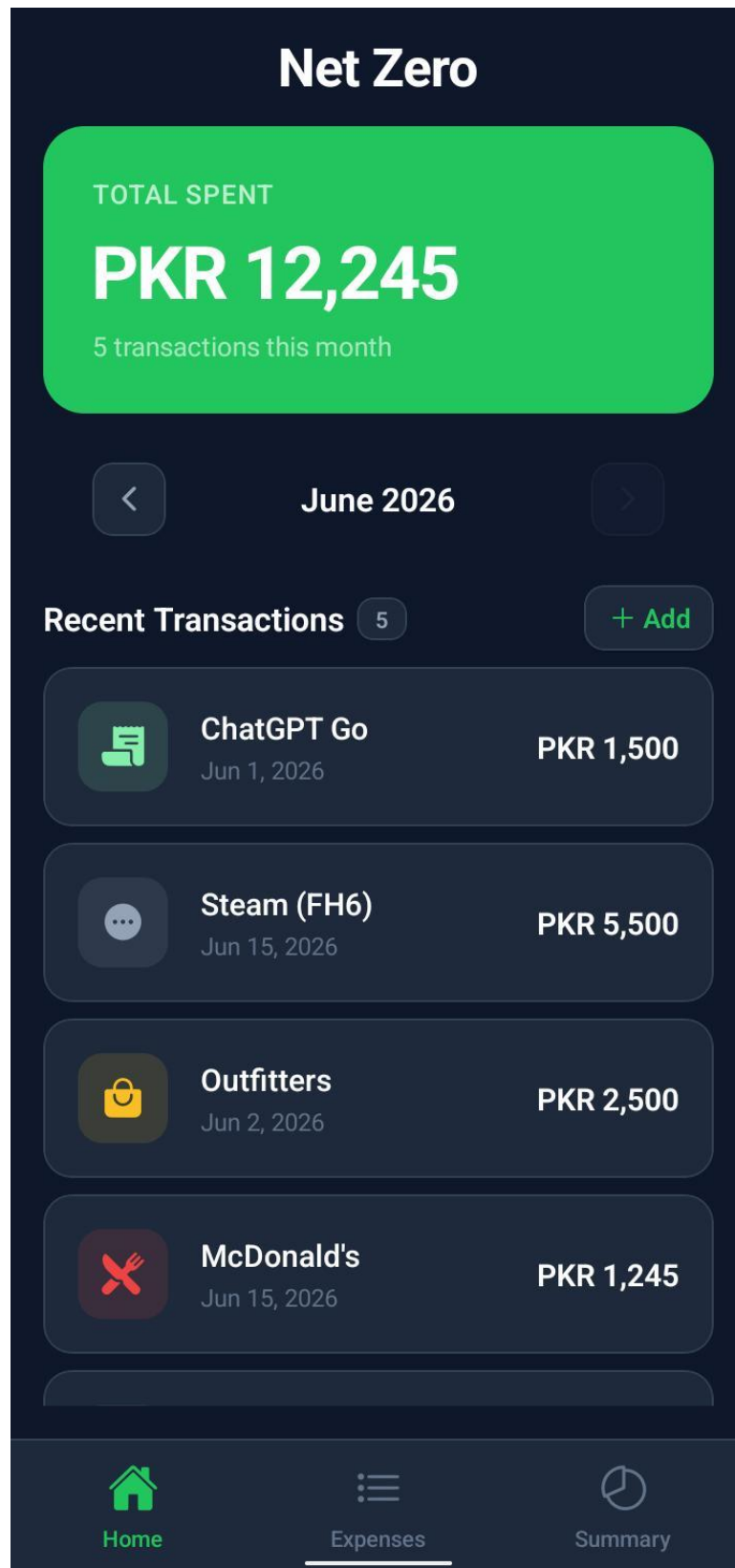
11. Conclusion

Net Zero is a simple but complete mobile expense tracker built with React Native and Expo. It covers all the basic needs of a user who wants to track their daily spending without any complicated setup.

The expected outcome is that any user can install the app and immediately start logging expenses. Over time they can use the monthly view and summary screen to see their spending patterns and manage their budget better.

Through this project we learned how to build a multi-screen React Native app, manage global state using Context API and `useReducer`, persist data locally with `AsyncStorage`, and integrate a charting library for data visualization. It also gave us good practice in organizing a React Native project with a proper folder structure.

10. Screenshots





Add Expense

Save

TITLE

e.g. Lunch at Bundu Khan

AMOUNT (PKR)

0

DATE

June 15, 2026



CATEGORY



Food



Transport



Shopping



Bills



Other

Expenses



June 2026



All

Food

Transport

Shopping

5 expenses



ChatGPT Go

Jun 1, 2026

PKR 1,500



Steam (FH6)

Jun 15, 2026

PKR 5,500



Outfitters

Jun 2, 2026

PKR 2,500



McDonald's

Jun 15, 2026

PKR 1,245



Petrol

Jun 13, 2026

PKR 1,500



Home



Expenses



Summary



Edit Expense

Save

TITLE

ChatGPT Go

AMOUNT (PKR)

1500

DATE

June 1, 2026



CATEGORY



Food



Transport



Shopping



Bills



Other

Summary



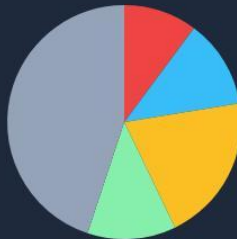
June 2026



Total this month

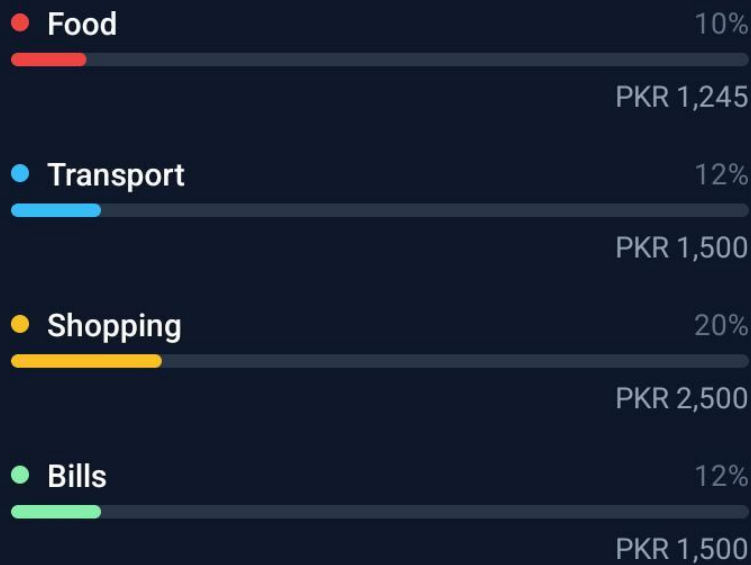
PKR 12,245

SPENDING BY CATEGORY



- 10% Food
- 12% Transport
- 20% Shopping
- 12% Bills
- 45% Other

Breakdown



Home



Expenses



Summary